Patrick O'Reilly

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First Class Honours CS student with two HPE internships building agentic AI services on Docker and Kubernetes. Seeking a graduate software engineering role from September 2027.

Education

University of Galway **Sept 2023 – May 2027**

B.Sc. Computer Science & Information Technology | First Class Honours (Years 1–3) *Galway, IE*

- **Relevant Coursework:** Advanced Programming (93%), Data Comms & Networking (92%), Programming Paradigms (90%)
- **Scholarships:** Excellence Scholarship (2023–24); University Scholar (2024–25) for exam results

Coláiste Éinde **Sept 2017 – June 2023**

Leaving Certificate | 625 points (maximum) *Salthill, Galway*

Technical Skills

Languages: Python, TypeScript, Java, C#, C, SQL
Web & Cloud: Docker, Kubernetes, AWS (Lambda, DynamoDB, S3), GCP, Spring Boot, FastAPI, Next.js, React, OAuth, CI/CD
AI & ML: Agentic AI, Multi-Agent Systems, LangGraph, Google ADK, MCP, RAG, Vector Search, Ollama, scikit-learn, LLM Evaluation, Langfuse
Data & Practices: PostgreSQL, DuckDB, TDD (PyTest), System Design, Regression Testing, Production Monitoring

Work Experience

Hewlett Packard Enterprise **Galway, IE**

Innovation Engineering AI Intern *(part-time from Sept 2026)* *Jan 2026 – Present*

Python, LangGraph, Google ADK, MCP, PostgreSQL, Docker, Kubernetes, OAuth, Langfuse

- Develop and demo AI prototypes to senior leadership, including CTOs, to help decide what gets built next. One made it onto the product roadmap within two months.
- Built an MCP retrieval service on PostgreSQL with vector schemas that serves context to downstream internal tools.
- Replaced a manual spreadsheet workflow with an AI tool that automates the analysis end-to-end.
- Containerised microservices with Docker on Kubernetes, secured service-to-service traffic with token-based OAuth, and run staging-to-production releases with regression tests and compliance checks.
- Upgraded an internal team website and recorded tutorials so a colleague could maintain it independently after handover.

Software Engineer Intern *June 2025 – Aug 2025*

Python, LangGraph, Kubernetes, RAG

- Built a LangGraph agent on my own that scans Kubernetes logs and diagnoses why workloads fail.
- Co-developed a chatbot that answers questions from HPE's internal documentation using RAG.

Projects

Taxi Chat – LLM Chatbot over 1.79B NYC Taxi Trips **Independent**

Python, DuckDB, Jev (TypeSafe), DeepSeek, FastAPI, React, deck.gl

- Built a chatbot over every NYC taxi and rideshare trip from 2019 to 2025 that declines questions it can't answer.
- On 46 questions it had never seen, answered 33 and got 30 right, and declined all 12 it couldn't answer; median reply 1.4 s.
- Jev, a classification model, picks the question type, place and dates with a confidence for each; code writes the SQL, the LLM only phrases the result, and a checker rejects any number not in the query result.

Local Multi-Agent Bug Diagnosis **Final Year Project – 2026/27**

Python, LangGraph, Ollama, Docker, tree-sitter

- Building a free bug-finding tool: a team of agents on a small open-weight model, running entirely on a laptop.
- Comparing a single local agent, a fixed specialist team, a dynamic orchestrator and Claude Code on ~20 Java and Python apps with planted bugs, scored by F1.

Ascend – AI-Powered Study Platform **Team of 3 Engineers**

Java, Spring Boot, AWS (Lambda, DynamoDB, S3), Google ADK, React, TypeScript

- Built 2 of 5 serverless microservices (notes and AI generation) plus their React frontend.
- Built a Google ADK agent that generates notes and quizzes from uploaded PDFs and answers questions about them in a chatbot.
- Designed S3-backed PDF storage with presigned URLs and PDFBox text extraction.

Chess.com MCP – MCP Server on npm **Independent**

TypeScript, MCP SDK, Node.js, Chess.com API

- Published an MCP server that lets Claude, Cursor or any MCP client answer "how did I do on Chess.com this month?"
- Summarises a month of games into a few hundred tokens of stats (record, rating, streaks, openings); npx install, no API key.

Personal Achievements

- **Academic:** maximum 625 Leaving Cert points; competed nationally in IMTA Senior Maths and ISTA Senior Science team events.
- **Public Speaking:** presented AI prototypes to HPE executive leadership (incl. CTOs); gave a university tech talk (lectured to 100+ students at ATU)
- **Chess:** Connacht team member on Chess.com. **Running:** completed the Great Limerick Run Half Marathon.